

Pipeline & Experiment Planning

Optical Camera Communication (OCC) Video Message Decoding — workflow set up for Google Colab + GitHub

1. Reproducible Pipeline Flowchart:

The proposed OCC video message decoding system follows a sequential pipeline in which the raw video is processed through several stages before the final message is reconstructed.

Overall Pipeline:

Raw Video → Ingest & Audit → ROI Detection → Intensity Signal Extraction → Preprocessing → Symbol Synchronization → Decode Bits → Evaluate → Reconstruct Message

➤ Raw Video

The process begins with the original OCC video stored in the raw/ directory. The raw videos are kept untouched and serve as the input for all experiments.

➤ Ingest and Audit

The video is first checked for basic properties such as:

- FPS
- Total number of frames
- Video integrity
- Resolution and other required properties

This ensures the video is suitable for further processing.

➤ ROI Detection

The **Region of Interest (ROI)** containing the transmitter signal is identified. Focusing on the ROI reduces unnecessary background information and directly affects the quality of the extracted intensity signal.

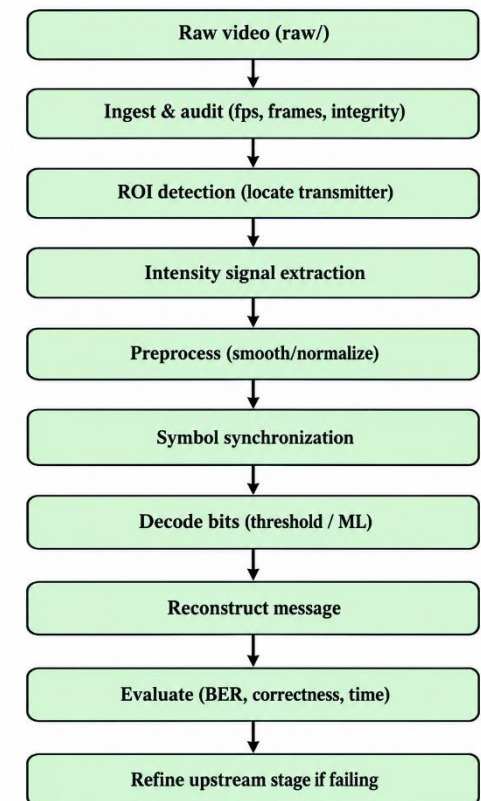
➤ Intensity Signal Extraction

The intensity within the selected ROI is extracted frame-by-frame to produce a time-varying intensity signal. This signal is used for synchronization and decoding.

➤ Preprocessing

The intensity signal is prepared for decoding using operations such as:

- Smoothing
- Normalization



The smoothing window and other parameters will be varied during experiments.

➤ **Symbol Synchronization**

Symbol synchronization identifies the timing and boundaries of transmitted symbols. Reliable synchronization is necessary for accurate bit decoding and will be evaluated across different videos.

➤ **Decode Bits**

The synchronized signal is converted into binary data using one of the planned approaches:

- Baseline threshold decoding
- Advanced signal-processing
- ML-assisted classification

This modular structure allows the methods to be compared under the same pipeline.

➤ **Evaluation**

The decoded output is evaluated using:

- Message correctness
- Bit-error rate (BER)
- Synchronization robustness
- Execution time per video

Poor results are used to identify which upstream stage requires improvement.

➤ **Feedback Loop**

The pipeline is iterative. Depending on the failure:

- **ROI problem** → revisit ROI detection
- **Signal problem** → revisit preprocessing
- **Timing problem** → revisit synchronization
- **Decoding problem** → revisit the decoding method
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2. Experiment Plan

➤ **Baseline Method**

The baseline method uses a fixed/adaptive threshold decoder applied directly to the extracted intensity signal without ML.

It serves as the mandatory reference for comparing all advanced methods. Key parameters include:

- Threshold type and value
- Smoothing configuration
- Symbol-timing assumption

All later methods will be evaluated against the baseline.

➤ Advanced Methods

A. Signal-Processing Pipeline

The advanced signal-processing approach uses:

- Adaptive local thresholding
- Smoothing
- Explicit symbol-timing recovery

Its performance will be compared with the baseline to determine whether improved signal processing increases decoding reliability.

B. ML-Assisted Symbol Classifier

The ML approach will use engineered features extracted from signal windows. Candidate classifiers are:

- KNN
- SVM
- Random Forest

The workflow is:

Signal Windows → Feature Extraction → Classification → Symbol/Bit Prediction → Message Reconstruction → Evaluation

The final model and feature set will be selected through experimentation.

➤ Project-Specific Experiment

Sensitivity / Ablation Study

The study will systematically vary:

1. ROI selection
2. Smoothing window
3. Threshold value
4. Symbol-timing assumption

These experiments will be conducted across multiple videos to determine which parameters have the greatest effect on decoding accuracy and robustness.

This study is planned as the Day 20 deep-dive, so data collection will begin early.

➤ Evaluation Criteria

Each method will be evaluated using four criteria:

1. Decoded-message correctness – whether the reconstructed message matches the reference.
2. Bit-error rate (BER) – bit-level decoding accuracy where ground truth is available.
3. Synchronization robustness – reliability of symbol timing across different videos and conditions.
4. Execution time – processing time per video for baseline, advanced, and ML methods.

These metrics will provide a consistent basis for comparing all decoding approaches.

3. Experiment Log Template

Every experiment will be recorded using a unique experiment ID.

The experiment log will contain the following information:

Exp. ID	Method / Variant	Parameters	Metrics	Notes / Next Action
1	Fixed-threshold baseline	threshold=128, no smoothing	Correctness, BER, sync, time	Struggles on low-light videos → try adaptive threshold
2	Adaptive threshold + smoothing	window=5, local mean threshold	Correctness, BER, sync, time	—
3	ML-assisted classifier	KNN, k=5, features=[...]	Correctness, BER, sync, time	—

Each experiment must represent **one complete run**.

Whenever a parameter is changed, a new experiment entry should be created.

Previous experiment results must not be overwritten.

The experiment log can be maintained using:

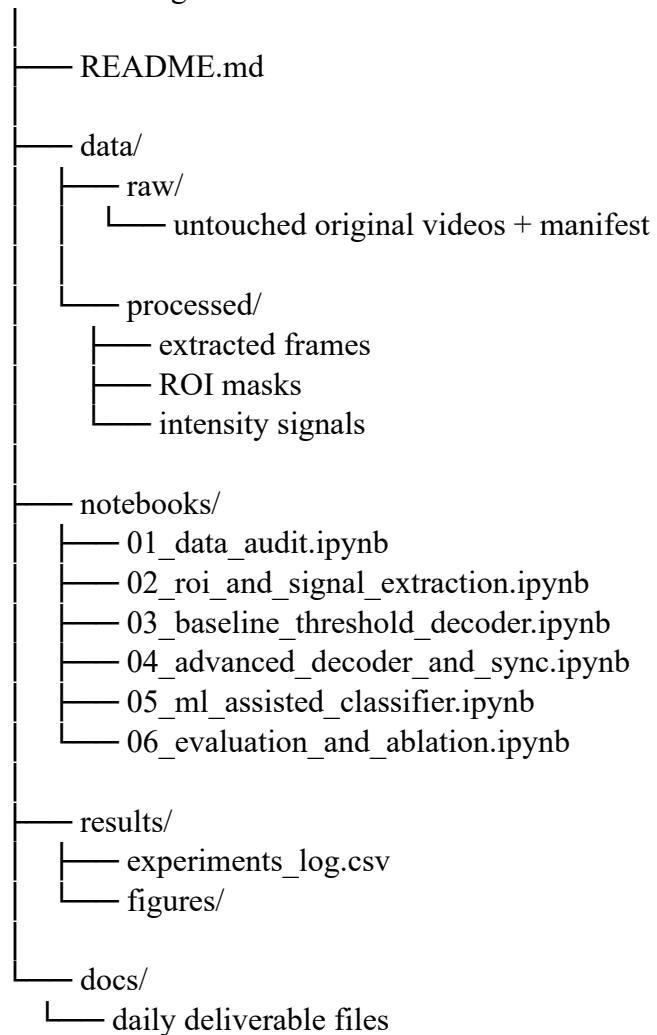
- experiments.csv
- An experiments section/cell within the relevant notebook

This ensures that every parameter change can be traced back to the corresponding result.

4. Repository / Folder Structure

The project repository will use the following structure:

TIH-Siri-Ettigowni/



5. Next Step:

- Reproducible OCC decoding pipeline
- Baseline decoding method

- Advanced signal-processing method
- ML-assisted classification approach
- Sensitivity/ablation study
- Evaluation criteria
- Experiment log
- Repository/folder structure

the project is ready to move to **Day 6**.

Day 6 will consolidate the **full literature + methodology + data-readiness package** in preparation for **Mentor Review 2**.

The Day 5 planning therefore provides the experimental foundation required for the next stage of the project.